SOIL EROSION AND CONSERVATION · NOTE ST04

Check Dams and Dam Site Selection — Earth and Composite Check Dams, Empirical Design and Investigations

Course topics: SWCE-501 T12, SWCE-503 T1, SWCE-503 T9, SWCE8.12 L8

Check dams

- A small dam of earth, masonry or concrete across a gully or stream that slows the flow, minimises channel scour and makes sediment settle. Small check dams are **gully plugs** — they slow and redistribute runoff, lengthen infiltration opportunity time, encourage silting, improve soil moisture and help plant cover establish.
- A series turns a uniform slope into steps with low drops, storing water and sediment, lowering the flow's silt-carrying capacity, adding groundwater, widening the channel and stopping it cutting down.
- **Main cause of failure:** an undersized spillway or notch — water overtops the earth portion, washes round the ends and the dam becomes useless. Estimate the peak flow from rainfall and watershed characteristics (rational, Curve Number or other methods).
- **Earth** is the cheapest and most available material — easy and economical, preferred wherever possible, and suited to small gullies (small earth dams get grassed ramps for overflow). Medium gullies need **composite check dams**: earth for the non-overflow part and brick or stone masonry, designed as a solid gravity dam, for the overflow part.

Small check dams in series

- A series flattens the bed and limits scour of the gully bed and sides at peak flow. Space them vertically so the top of each downstream dam is about level with the base of the one above — forming flat terraces over which water flows without gaining much velocity. Stream gradients up to **1 in 8** can be controlled economically.
- **Heights:** spillway level ≈ 1 m above ground, plus ≈ 30 cm for the spillway and ≈ 15 cm freeboard — an ordinary check dam stands ≈ 145 cm above ground, depending on the gorge section.

Base width: $b = H/\sqrt{\rho} + d$ H = height to notch crest; d = maximum depth over crest; ρ = specific gravity of masonry (2.25 g/cc, or 2200 kg/m^3)

Crest width (as printed): $T = 3d/(2B)$ B = base width

Notch length ≥ 7/8 of the average bed width of the gully

Apron length = L + ½d L = width of notch

Horizontal floor length = $2d + \sqrt{(d \cdot h)}$ h = difference between water levels above and below the weir

Depth of water cushion (as printed) = $(d/2) \cdot \sqrt{(h - 2)}$

Side revetment length = $4(d + h)$, minimum 6 m from the edge of the solid apron

These are empirical rules quoted from the textbook. Two need checking before use: "T = 3d/(2B)" gives a ratio rather than a width, and "(d/2)√(h - 2)" is undefined for h < 2. Verify them against the source text.

- Revetments protect the gully sides from eddies; pitch the bed with stone over half the revetment length. Side slopes 1 : 1 to 1.5 : 1; stone about 30 cm thick.

Major check dams

- Masonry, concrete and earth dams for soil conservation, ravine reclamation, runoff retention and flow regulation need proper investigation, planning, design and construction for efficiency and economy.
- **Preliminary investigations** (reconnaissance, secondary data, outline surveys) compare sites and justify the cost; with limited data, conservative assumptions lead to over-design, whereas better data save money.
- **Land survey plan:** preliminary route surveys, bench marks, channel grades and cross-sections, with levels of buildings, roads and structures. **Secondary data:** maps (Surveyor General of India), soils (state and central soil surveys), weather (Meteorological Department). **Preliminary boring** for foundation stability and fill material — a major factor in site choice.
- **Final surveys:** channel profiles extending below the structure to a stable grade and above it to the limit of its influence — at least 150 m each way; cross-sections for orientation and fill estimates; a reservoir survey of the whole impounded area; watershed maps from aerial photographs; soil borings along the sections to determine strata.
- **Plans** show all works and quantities, the hydrologic, hydraulic, structural, foundation and embankment designs, and drawings (profiles, cross-sections, watershed and site maps, structural drawings, contour and location maps, longitudinal sections).

Selecting a dam site

Dam, spillway and reservoir should each get optimum conditions.

1. Avoid unstable ground — springs, seeps, old landslides — and rock obstructions (boulders, outcrops).
2. Foundation: hard, groutable rock without faults (essential for gravity and arch dams); earth dams can be built on most foundations, but watch erosion, settlement and seepage, which depend on stratification, permeability and homogeneity.
3. Keep the dam as short as possible — a narrow gorge or canyon saves fill and cost.
4. Keep the dam and reservoir lower than the surrounding farmland to limit submergence; command the area by gravity (pumping is expensive).
5. Have medium-textured soil nearby — clay cracks on drying and sand does not hold water; ensure road access for materials and equipment.
6. Upstream topography should give enough storage; consider the danger to life and property if the dam fails; study watershed runoff (diversion from adjacent streams may help).
7. Do not site the reservoir on highly permeable material — a fine-textured bottom is ideal.
8. Spillway: prefer a wide, shallow spillway in non-erodible material to a deep, narrow one; the emergency spillway usually goes round one end, where its flow will not wash the dam. Steep canyons need concrete or masonry chutes or pipe drop spillways.
9. Control erosion in the watershed to prevent reservoir silting.
10. Environmental impacts are usually small — fish deaths in dry periods, loss of fodder area, wildlife effects, mosquito breeding and spread of disease.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The main cause of check-dam failure is

(a) poor stone
(b) insufficient spillway or notch capacity
(c) too much vegetation
(d) seepage

Q2. Stream gradients that can be controlled economically by check dams are up to

(a) 1 in 2
(b) 1 in 8
(c) 1 in 50
(d) 1 in 100

Q3. An ordinary small check dam stands about ___ above ground

(a) 45 cm
(b) 1 m
(c) 145 cm
(d) 3 m

Q4. Notch length should be at least ___ of average gully bed width

(a) 1/2
(b) 3/4
(c) 7/8
(d) 1.5 times

Q5. Side revetment length is 4(d + h) with a minimum of

(a) 2 m
(b) 6 m
(c) 10 m
(d) 15 m

Q6. Channel profiles for major check dams should extend at least ___ each way

(a) 15 m
(b) 50 m
(c) 150 m
(d) 1 km

Q7. A composite check dam has

(a) all earth
(b) earth non-overflow and masonry overflow sections
(c) all concrete
(d) gabions only

Q8. The best soil near a dam site for construction is

(a) clay
(b) sand
(c) medium-textured soil
(d) gravel

Answers

Q1 — (b) insufficient spillway or notch capacity
Q2 — (b) 1 in 8
Q3 — (c) 145 cm — *1 m + 30 cm spillway + 15 cm freeboard.*
Q4 — (c) 7/8
Q5 — (b) 6 m
Q6 — (c) 150 m
Q7 — (b) earth non-overflow and masonry overflow sections
Q8 — (c) medium-textured soil